

Structure of Any C++ Program

The General Format Of Any Program Written In C++ Is :

```
#include<iostream>
using namespace std;
int main()
{
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    return 0;
}
```

Built in Library

1- #include<iostream>

It helps the user (programmer) to input data for the program during execution from the keyboard and print the results on the execution screen.

2- Std

A library containing the commands, functions, and data types we need to write any program, such as:

- ✚ Input commands (**Cin**) and output commands (**Cout**) (printing).
- ✚ Variables types such as **Int**, **Char**, **String**, **Float**.

Basic Printing Principles

When using the **Cout** command, observe the following principles:

- To display a number, leave it as is.
- To display the value of a variable, leave it as is.
- To display a character, place it between ' '
- To display a word or text, place it between . " "

Notes

A semicolon (;) is placed at the end of each line of code to indicate the end of that line.

- Any text enclosed in double quotes (" ") is printed exactly as is.

Example (1) : use Cout command to print the text "Welcome to C++" on the run screen .

```
#include<iostream>
using namespace std;
int main()
{
cout<<"Welcome to C++";
return 0;
}
```

- ❖ If we want to print another text in the next line on the screen we will add the command "endl" which means newline. So the program will be as following.

```
#include<iostream>
using namespace std;
int main()
{
cout<<"Welcome to C++"<<endl;

cout<<"welcome to the first stage students "<<endl;

return 0;
}
```

✚ Also we can use the command `"\n"` which give the same previous effect. This command is abbreviation to new line.

✚ To Add spaces between words we use the command `"/t"` which is abbreviation to Tab Space.

Example (2): using the learned commands in this laboratory draw the shape bellow.

```

**      **
**      **
*****
**      **
**      **
```

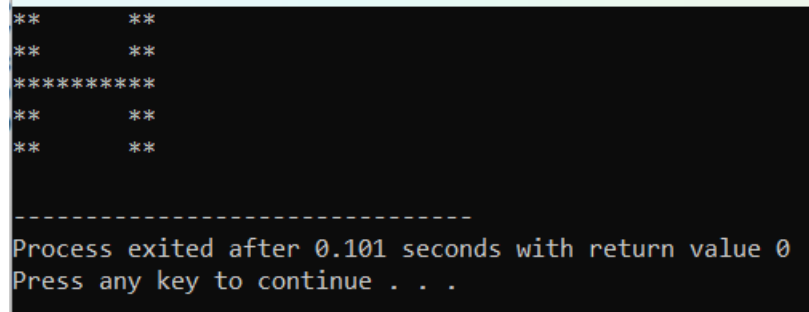
Sol.

```
#include<iostream>
using namespace std;
```

```
int main()
{
cout<<"**"<<"\t"<<"**"<<endl;
cout<<"**"<<"\t"<<"**"<<endl;
cout<<"**"<<"**"<<"**"<<"**"<<"**"<<endl;
cout<<"**"<<"\t"<<"**"<<endl;
cout<<"**"<<"\t"<<"**"<<endl;

return 0;
}
```

The result of program running is :

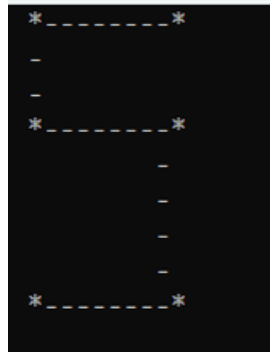


```
**      **
**      **
*****
**      **
**      **

-----
Process exited after 0.101 seconds with return value 0
Press any key to continue . . .
```

LABORATORY TASKS:GROUP (A)

using the learned commands in this laboratory draw the shape bellow.
Using “*” and “-” symbols.

GROUP (B)

using the learned commands in this laboratory draw the shape bellow.
Using “*” symbol.

